

Project Documentation: Cost-Optimized Daily AWS Glue ETL Job

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Problem Scenario

As a data engineer, you are tasked with building a cost-effective and automated ETL pipeline to process employee data stored in Amazon S3. The requirement is to:

- Extract raw employee data from an S3 bucket.
 - Transform it by selecting only relevant columns.
 - Store the cleaned output in another S3 location.
 - Schedule it to run daily, **without high operational cost**, and ensure scalability for future enhancements.
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Project Goal

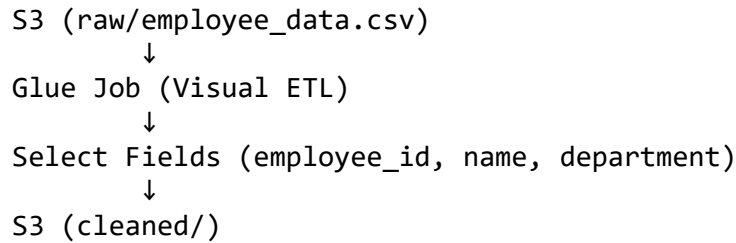
Design a **serverless, cost-efficient ETL pipeline using AWS Glue Studio** that:

1. Reads a raw CSV file from an S3 bucket.
 2. Selects only necessary columns (employee_id, name, department).
 3. Writes the output to a target S3 location.
 4. Runs daily via a scheduled trigger using low-cost configuration.
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Tools & Services Used

AWS Service	Purpose
Amazon S3	Store raw input and cleaned output files
AWS Glue Studio	Visual ETL job for transformation
AWS Glue Triggers	Schedule daily ETL execution
IAM	Assign access permissions to Glue jobs

Solution Architecture



Step-by-Step Approach

☒ Step 1: S3 Bucket Setup

- Created a bucket with folders:
 - raw/: for uploading employee_data.csv
 - cleaned/: for storing transformed data

☒ Step 2: IAM Role Configuration

- Created IAM role: glue-flex-etl-role
- Attached policies:
 - AWSGlueServiceRole
 - AmazonS3FullAccess (*fine-tune later for production*)

☒ Step 3: Build Glue ETL Job in Glue Studio

- Started a Visual Job from blank graph.
- Configured:
 - Source Node: Read from s3://your-bucket/raw/employee_data.csv
 - Transform Node: Select only employee_id, name, department
 - Target Node: Write to s3://your-bucket/cleaned/

☒ Step 4: Optimize Glue Job for Cost

- Selected:
 - **Execution Class:** FLEX (spot-like savings)
 - **Glue Version:** 4.0 (latest features)
 - **IAM Role:** glue-flex-etl-role

☒ Step 5: Manual Test Run

- Triggered the job manually
- Confirmed output landed correctly in cleaned/ folder with selected columns only

☑ Step 6: Automate with Trigger

- Created a daily schedule:

```
cron(0 1 * * ? *) // Executes daily at 1:00 AM UTC
```
- Enabled trigger and attached it to Glue Job

Input File Sample: `employee_data.csv`

employee_id	name	department	salary	joining_date
101	Alice	HR	50000	2020-01-15
102	Bob	Engineering	75000	2019-03-22

Output File Stored in:

`s3://your-bucket/cleaned/`

With selected fields only:

employee_id	name	department
101	Alice	HR
102	Bob	Engineering

Cost Optimization Justification

Optimization Area	Description
Execution Class: FLEX	Up to 34% cheaper compared to standard Glue jobs
Scheduled Daily Run	Prevents idle compute time or manual intervention
Fully Serverless	No infrastructure to maintain or scale manually

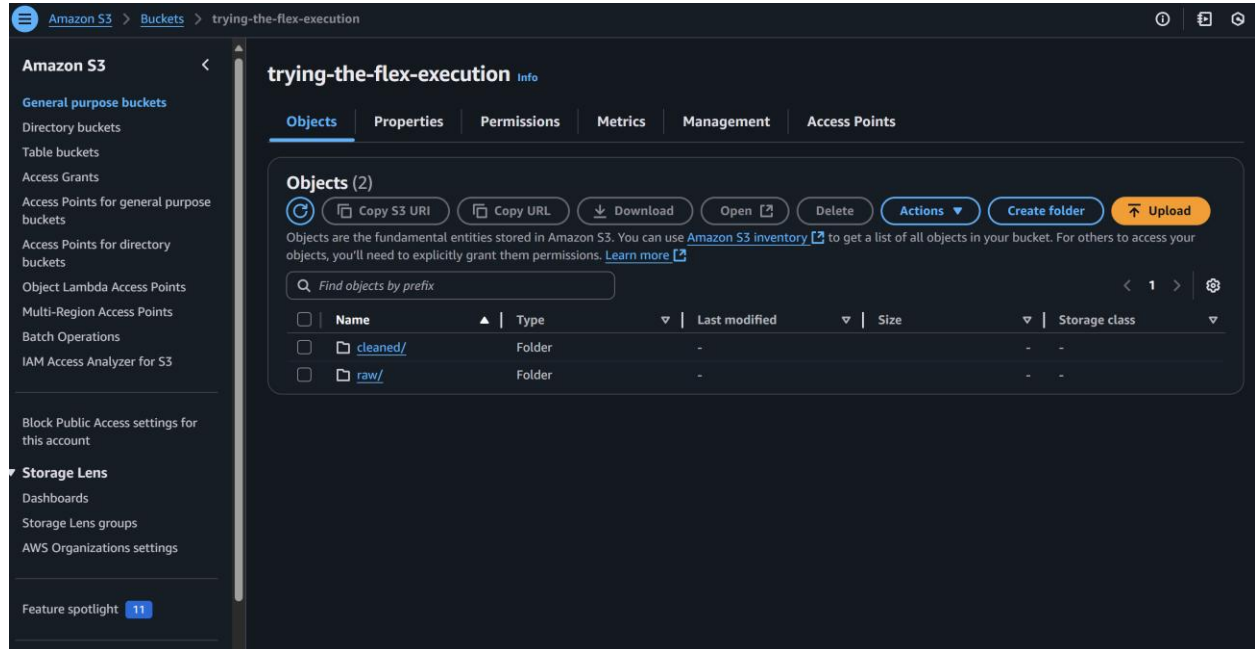
Learning Outcomes

- Gained hands-on experience with AWS Glue Studio (Visual interface)
 - Understood cost-saving via FLEX execution class
 - Learned how to automate ETL jobs with daily triggers
 - Practiced clean data transformation using S3 and Glue together
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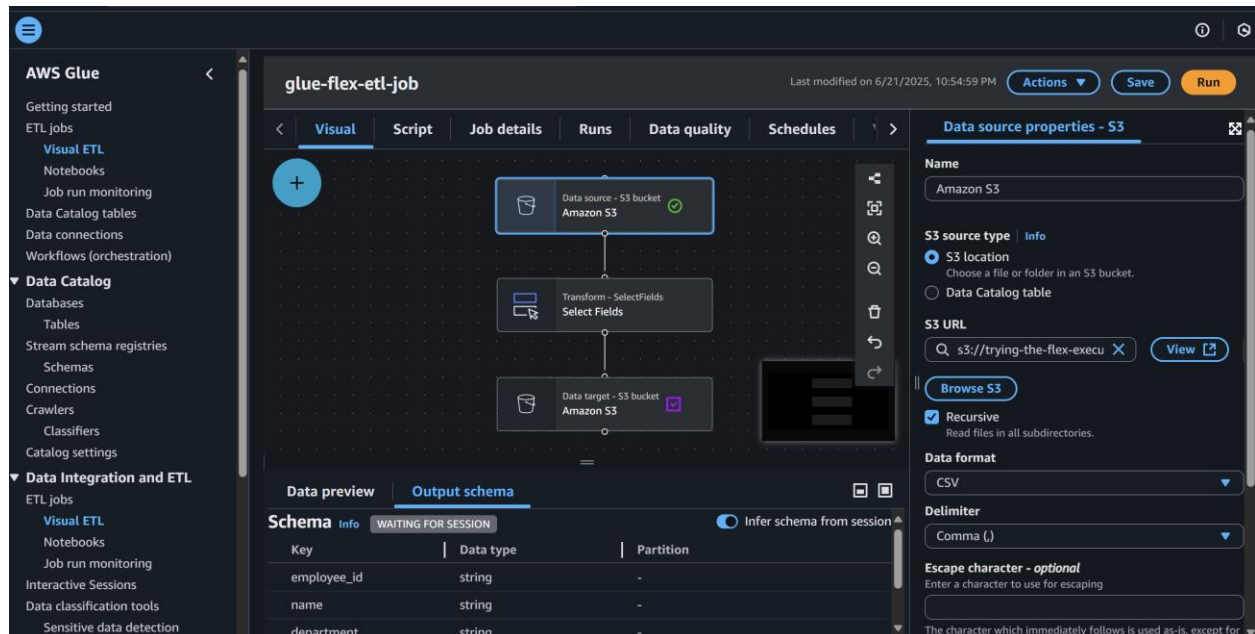
Paste Workspace Outputs Here

Paste your screenshots, logs, or Athena query results here:

[S3 Folder Structure Screenshot]



[Glue Job Run Screenshot]



[Cleaned CSV Output Sample]

	A	B	C	D	E	F	G
1	employee_ name		department				
2	101	Alice	HR				
3	102	Bob	Engineering				
4	103	Charlie	Marketing				
5	104	David	Engineering				
6	105	Eve	HR				

[Trigger Configuration Screenshot]

AWS Glue

Getting started

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Data Catalog

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

Data Integration and ETL

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Interactive Sessions

Data classification tools

Sensitive data detection

glue-flex-etl-job

Last modified on 6/21/2025, 10:54:59 PM

Actions

Save

Run

Visual

Script

Job details

Runs

Data quality

Schedules

Version Control

Job runs (1/1)

Info

Last updated (UTC)

June 22, 2025 at 01:27:06

View details

Stop job run

Troubleshoot with AI

Table View

Card View

Filter job runs by property

Run status

Retries

Start time (Local)

End time (Local)

Duration

Capacity...

Worker type

Succeeded

0

06/21/2025 22:55:36

06/21/2025 22:57:56

2 m 5 s

10 DPU's

G.1X

Run details

Input arguments (9)

Logs

Run insights

Metrics

Troubleshooting analysis - preview

Spark UI

Job name

glue-flex-etl-job

Start time (Local)

06/21/2025 22:55:36

DPU Hours

0.186111

Glue version

4.0

Last modified on (Local)

06/21/2025 22:57:56

Id

jr_50a80cb21100cba73fbc3b2f9b2792

End time (Local)

06/21/2025 22:57:56

Worker type

G.1X

ba0235824320ae87a1fa37165cf6ab55d



#AWSGlue #S3ETL #GlueStudio #CostOptimization #FLEXExecution #DailyAutomation #ServerlessETL